

VideoThinker: Building Agentic VideoLLMs with LLM-Guided Tool Reasoning

Chenglin Li^{1,5}, Qianglong Chen¹, Feng Han^{2,5}, Yikun Wang^{2,5}, Xingxi Yin¹, Yan Gong¹, Ruilin Li^{3,5},
Yin Zhang^{1*}, Jiaqi Wang^{4,5*}

¹Zhejiang University, ²Fudan University, ³Wuhan University,
⁴Shanghai AI Lab, ⁵Shanghai Innovation Institute

Abstract

Long-form video understanding remains a fundamental challenge for current Video Large Language Models (VideoLLMs). Most existing models rely on static reasoning over uniformly sampled frames, which weakens temporal localization and leads to substantial information loss in long videos. Agentic tools, such as temporal retrieval, spatial zoom, and temporal zoom, offer a natural way to overcome these limitations by enabling adaptive exploration of key moments. However, constructing agentic video-understanding data requires models that already possess strong long-video comprehension, creating a circular dependency. We address this challenge with VideoThinker, an agentic VideoLLM trained entirely on synthetic tool-interaction trajectories. Our key idea is to convert videos into rich captions and employ a powerful agentic LLM to generate multi-step tool-use sequences in the caption space. These trajectories are subsequently grounded back to video by replacing captions with actual frames, yielding a large-scale interleaved video-tool reasoning dataset without requiring any long-form understanding from the underlying VideoLLM. Training on this synthetic agentic dataset equips VideoThinker with dynamic reasoning capabilities, adaptive temporal exploration, and multi-step tool use. Remarkably, VideoThinker significantly outperforms both caption-only LLM agents and strong VideoLLM baselines across long-video benchmarks, demonstrating the effectiveness of tool-augmented synthetic data and adaptive retrieval-zoom reasoning for long-form video understanding.

1. Introduction

Video understanding is a fundamental task that requires reasoning over both spatial and temporal dimensions of videos [3, 15, 18, 22, 41, 44, 58]. Recent advances in Video Large Language Models (VideoLLMs) have achieved remarkable progress in this area [1, 14, 17, 19, 48, 53].

Despite these advances, existing models still struggle with long-form videos [9, 32, 37], where relevant visual evidence is sparse, distributed across time. Static reasoning over uniformly sampled frames often causes major information loss and poor temporal localization, while also being computationally costly for long videos with hundreds of frames.

A promising avenue for improving long-video understanding lies in the concept of interleaved multimodal reasoning, which involves combining visual perception with language reasoning in a dynamic and adaptive manner [13, 35, 56]. Inspired by successful techniques in the image domain, such as those seen in OpenAI’s o3 model [23], concurrent work attempts to extend such interleaved reasoning to videos, proposing the notion of *thinking with videos* [10, 21, 49]. However, these approaches face key limitations: (1) they often rely on proprietary models such as Gemini-Pro [5] to construct agentic video-understanding data, which requires extensive prompt engineering and multi-stage filtering; Moreover, although Gemini-Pro can produce interleaved video-reasoning traces, it still lacks the *emergent* video-interleaved reasoning capability observed in o3’s image reasoning [23]; and (2) they primarily perform single temporal zoom operations without a temporal retrieval mechanism, making them inefficient at identifying key temporal intervals. In contrast, open-source LLMs such as Qwen3 [40] and DeepSeek [20] exhibit strong tool-augmented reasoning capabilities as language agents for long-form video understanding. They employ various tools, with VideoLLMs generating intermediate frame captions as tools to support video reasoning [7, 33, 36, 46]. However, in such frameworks, the VideoLLM serves merely as a passive captioning module, while the LLM itself cannot directly perceive visual information.

Motivated by these insights, we propose **VideoThinker**, an agentic VideoLLM trained entirely on synthetic tool-interaction trajectories. The core idea is to first convert videos into rich textual captions and then leverage a powerful agentic LLM to generate multi-step tool-use sequences within this caption space. To enable effective long-video reasoning, we design two complementary agentic tools: (1)

*Corresponding Authors

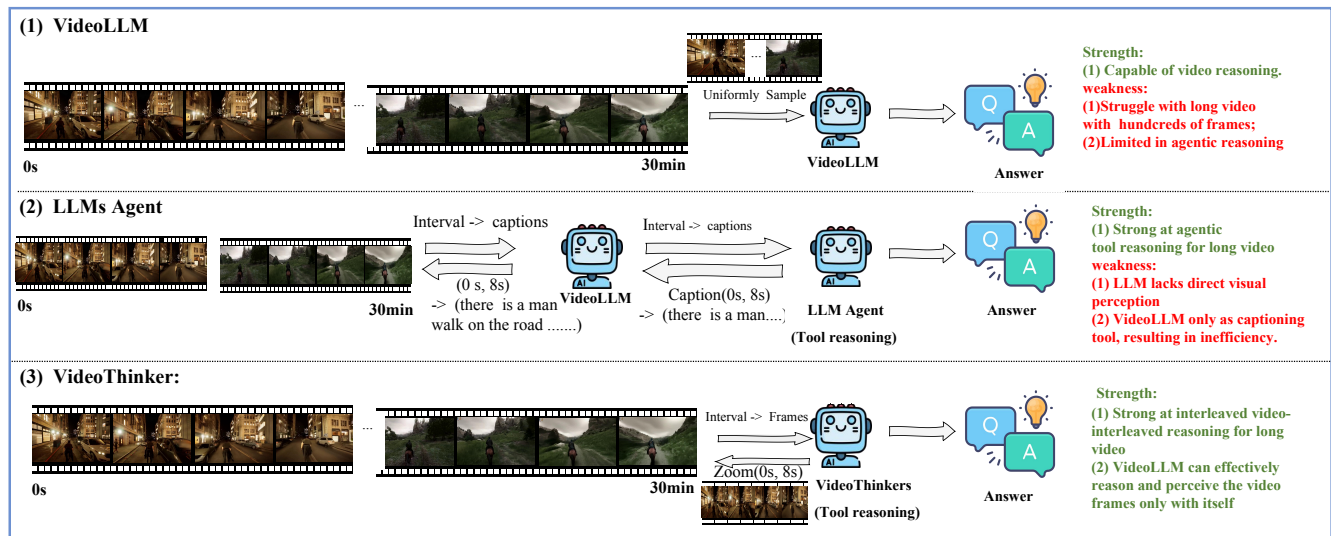


Figure 1. Comparison of VideoThinker with VideoLLMs and LLM agents. VideoThinker excels at interleaved video reasoning on long videos, using agentic tools to iteratively perceive and reason over video frames step by step.

Temporal Retrieval, which identifies candidate temporal intervals that may contain relevant information by leveraging audio transcripts (as subtitles), scene descriptions, and subtitle-based summaries; and (2) **Temporal Zoom**, which inspects intervals at a finer granularity through more detailed subtitles or frames. By leveraging these tools together with the LLM’s tool-augmented reasoning capability, we construct multi-turn tool-interaction trajectories. When the LLM invokes the temporal-zoom tool, the VideoLLM generates intermediate textual captions, which are subsequently replaced with actual video frames to form fully video-interleaved reasoning data. These trajectories are then used to fine-tune the VideoLLM, enabling it to actively retrieve and perceive key frames during reasoning, effectively bridging the gap from text- and image-based reasoning to true video reasoning. Furthermore, VideoThinker incorporates a confidence-gated tool controller, achieving strong performance gains: +6.8% on MLVU and +10.6% on LVBench over vanilla VideoLLMs, and +3.9% and +3.5% over caption-only LLM agents equipped with our tools. Our main contributions are summarized as follows:

- We construct a high-quality agentic video–tool reasoning dataset guided by LLM-based tool reasoning. This synthetic data captures diverse temporal reasoning and tool-use behaviors essential for long video understanding.
- We equip **VideoThinker** with dynamic reasoning, adaptive temporal exploration, and multi-step tool use via *Temporal Retrieval* and *Temporal Zoom* for efficient long-video understanding.
- Extensive experiments demonstrate that **VideoThinker**

outperforms strong VideoLLM and LLM-agent baselines across long-video benchmarks, validating the effectiveness of tool-augmented synthetic data and adaptive retrieval–zoom reasoning.

2. Related Work

2.1. Agentic LLMs for Video Understanding

Recent advances in large language models (LLMs), particularly their reasoning and planning capabilities, have spurred the development of agents [11, 43]. By leveraging external tools [24, 47], LLMs can acquire information, plan, and execute actions in complex environments, bridging the gap between general-purpose language models and real-world applications. Extending this paradigm to video understanding, recent works integrate LLM reasoning with agentic tools to tackle complex video tasks without costly training. However, applying such agents to video understanding poses challenges, as many existing tools do not generalize well across temporal sequences. Several works have proposed video-specific adaptations of the LLM-agent paradigm [7, 16, 33, 46]. LifeLong-Memory [34] constructs a text-based episodic memory from video narrations, enabling LLMs to reason and retrieve relevant information for downstream tasks. DoraemonGPT [42] introduces a MCTS-based prompting strategy, combining tool use with structured memory for video reasoning. VideoAgent [33] frames video understanding as a sequential decision-making process, where the agent decides whether to gather more information or produce an answer, mimicking human video interpretation strategies.

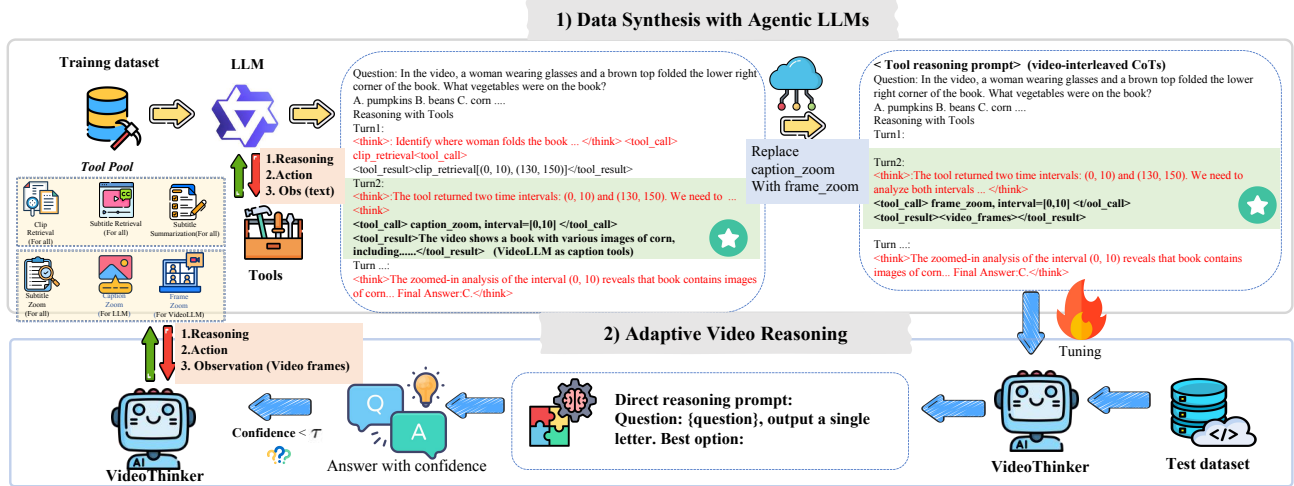


Figure 2. VideoThinker integrates retrieval and zoom tools for multi-turn reasoning. LLMs use **caption_zoom** to generate reasoning data from videos, which is later replaced by **frame_zoom** and video frames to build agentic video understanding CoTs.

VideoTree [36] enhances efficiency and interpretability by adaptively extracting key visual information in a coarse-to-fine manner and organizing it in a human-readable tree for LLM reasoning. Related approaches, such as DVD [52] and VideoExplorer [45], employ agentic search strategies over segmented video clips to tackle long-form VideoQA. Together, these works demonstrate the potential of agentic LLMs for structured, tool-augmented reasoning in complex video understanding tasks. A key limitation of these approaches is that they rely on a strong LLM as the core reasoning engine, while VideoLLMs serve only as auxiliary tools (e.g., generating captions). In contrast, our method elevates the VideoLLM itself to the reasoning core, effectively simplifying the agent architecture. By combining the capabilities of a VideoLLM with the agentic planning paradigm, our framework achieves better performance and reduces reliance on LLMs, improves efficiency, and streamlines the integration of multimodal reasoning for long-form video understanding.

2.2. Multimodal reasoning with MLLMs

Multimodal reasoning differs from pure text reasoning in that it enables models to directly manipulate visual content, such as zooming into specific image regions [6, 13, 27, 30, 31, 35, 54, 56]. Leveraging this capability, our work focuses on deciding which video clips to zoom in on, enabling efficient processing of long videos. Recent concurrent works, such as Open-o3-Video [21] and Video-MTR [39], also adopt dynamic processing strategies using chain-of-thought reasoning (CoT) over video content. These approaches generate multimodal tool data for training advanced VideoLLMs with Gemini-2.5-pro. On one hand, Gemini-pro it-

self has not demonstrated “thinking with videos” capabilities via multimodal tools (contrary to O3, which shows “thinking with images” on image tasks). On the other hand, synthesizing high-quality multi-turn video reasoning data typically requires data validation and extensive data engineering, making direct generation costly. Differently, we construct the tool-augmented reasoning ability from the language agents and use video frames captions as textual proxies for multimodal tool outputs. This approach allows efficient data synthesis while leveraging open-source models such as Qwen3 to produce high-quality video-interleaved CoT. In this way, our framework efficiently equips VideoLLMs with the capability to “think with videos” through tool-augmented reasoning, without heavy computational or annotation costs.

2.3. Long-form VideoQA and Video-LLMs

Long-form VideoQA requires reasoning over extended temporal contexts and capturing causal dependencies across events [9, 32, 37, 38, 58]. Recent Video-LLMs extend temporal encoders for joint spatio-temporal reasoning [1, 17], but still face memory and computation bottlenecks when processing long videos. To alleviate this, several methods convert videos into textual captions or keyframe summarization, allowing LLMs to perform text-based reasoning [33, 36]. While effective for scalability, these approaches lose fine-grained temporal details [50, 51]. Motivated by these limitations, we propose VideoThinker, which focuses on key segments and performs frame-level video-interleaved reasoning, enabling efficient long-form video understanding.

3. Method

In this section, we present VideoThinker, an agentic VideoLLM for long-form video understanding through dynamic, tool-augmented reasoning. To tackle temporal sparsity and information loss, we design two complementary tools—Temporal Retrieval and Temporal Zoom—that enable adaptive exploration of key moments. We then construct a synthetic training dataset based on tool-interaction trajectories from LLM, allowing the VideoLLM to learn multi-step reasoning and video-frame perception.

Describe any visual evidence in the video that directly helps answer the question. If there is no relevant evidence, provide an objective summary of the video content.
Question: <question>

Figure 3. The prompt designed to enable VideoLLM to serve as a caption generation tool.

3.1. Tools Design

We design two complementary agentic tools for efficient retrieval and reasoning over long-form videos. It includes functional groups: Temporal Retrieval tools (Clip Retrieval, Subtitle Retrieval, Subtitle Summary) for multimodal content access, and Temporal Zoom tools (Frame Zoom, Subtitle Zoom) for fine-grained inspection of some intervals. This modular design enables adaptive focus on key segments while preserving global context. Under an agentic search paradigm, the agent decomposes queries, composes tool calls, and iteratively gathers evidence to refine understanding and localize relevant information within long videos.

3.1.1. Temporal Retrieval

ClipRetrieval This tool provides coarse-grained access to semantically relevant temporal regions in long videos. Given a `video_path` and a text query, the video is first segmented into 10-second clips, each encoded using LanguageBind-Video [59] to obtain clip-level embeddings. The tool then retrieves the top-ranked clips with the highest semantic similarity to the input query and returns their corresponding temporal intervals. By iteratively invoking `ClipRetrieval` with refined queries informed by intermediate reasoning results, the agent can progressively zoom in on the most relevant video segments, efficiently narrowing the search space for subsequent analysis. **cvpr** **SubtitleRetrieval** This tool enables fine-grained text-level

retrieval over automatically transcribed subtitles. Given a `video_path` and a text query, Whisper [25] is employed

to transcribe the video audio stream, after which the tool retrieves subtitle segments most semantically relevant to the query, along with their timestamps. Through iterative refinement, `SubtitleRetrieval` allows the agent to align textual evidence with visual cues, thereby precisely localizing key temporal intervals.

SubtitleSummary This tool generates concise, query-focused summaries to support global video understanding. Built upon Qwen3-30B [40], it processes the complete subtitle transcript to extract information most relevant to the input query. The resulting summaries enable the agent to efficiently grasp the overall narrative flow and contextual dependencies of long-form videos, serving as high-level semantic guidance for downstream reasoning.

3.1.2. Temporal Zoom

FrameZoom This tool supports fine-grained visual inspection by extracting raw frames within a specified temporal interval. Given start and end timestamps, it retrieves the corresponding frames for detailed analysis. For example, if a video contains 32 frames in total but only 2 fall within the interval $[0, 10]$ seconds, invoking `FrameZoom(0, 10)` re-samples this interval to return 8 frames, thereby increasing visual density and improving perceptual detail for localized reasoning.

SubtitleZoom This tool extracts subtitle segments corresponding to a specified temporal interval. By providing start and end timestamps, the agent obtains localized subtitle text aligned with the chosen segment, enabling fine-grained multimodal reasoning within temporally bounded contexts.

CaptionZoom This tool acts as a semantic bridge between visual frames and text-based reasoning. It first invokes `FrameZoom` to extract frames from a given interval, then employs a VideoLLM to generate natural-language captions summarizing key visual events, objects, and interactions. During tool-reasoning data synthesis, the LLM agent queries `CaptionZoom` for localized visual semantics using textual descriptions rather than raw frames, effectively simulating perceptual grounding through tool-based interaction. In the final training stage, these generated captions are replaced by visual frame tokens, enabling the VideoLLM to internalize structured reasoning patterns grounded directly in visual representations.

3.2. Data Synthesis with Agentic LLMs

Given a training dataset $\mathcal{D} = \{(v_i, x_i, y_i)\}_{i=1}^M$, where v_i denotes a video, x_i is the corresponding question, and y_i is the ground-truth answer, we construct structured tool system prompts to elicit tool-based reasoning from the LLM. For each query x_i , a video caption c_i of v_i is generated using the VideoLLM. The query x_i , caption c_i , and tool system prompt template p are then combined to form the model

Algorithm 1 Tool-Reasoning Data Synthesis

Require: Training sample (v_i, x_i) , max steps T , LLM P , VideoLLM, tool set $\mathcal{M}$, action space $\mathcal{A} = \mathcal{M} \cup \{\text{Answer}\}$

- 1: Generate video caption: $c_i \leftarrow \text{VideoLLM}(v_i)$
- 2: Construct initial input prompt: $p_i \leftarrow p(x_i, c_i)$
- 3: Initialize reasoning history: $\mathcal{H}_0 \leftarrow \{p_i, \mathcal{A}\}$
- 4: **for** $t = 1$ to T **do**
- 5: Generate reasoning step: $\hat{r}_t \leftarrow P.\text{reason}(\mathcal{H}_{t-1})$
- 6: Select action: $(\alpha_t, \hat{p}_t) \leftarrow P.\text{call}(\hat{r}_t, \mathcal{H}_{t-1})$
- 7: **if** $\alpha_t = \text{Answer}$ **then**
- 8: **break**
- 9: **end if**
- 10: Invoke tool: $\hat{\omega}_t \leftarrow \alpha_t(\hat{p}_t)$
- 11: Update history: $\mathcal{H}_t \leftarrow \mathcal{H}_{t-1} \cup \{(\hat{r}_t, \alpha_t, \hat{\omega}_t)\}$
- 12: **if** $t = T$ **then**
- 13: Produce final answer: $\hat{y}_i \leftarrow P.\text{answer}(\mathcal{H}_t)$
- 14: **end if**
- 15: **end for**
- 16: **return** $\hat{y}_i, \mathcal{H}$

input. The LLM P processes this input to produce both the reasoning trajectory $\hat{r}_i$ and a predicted answer $\hat{y}_i$. Each trajectory $\hat{r}_i$ represents a multi-turn reasoning path, where each step corresponds to a reasoning decision and a tool invocation. During data synthesis, CaptionZoom is the only visual access point: it converts frames from a chosen interval into temporally grounded captions, which the LLM then reasons over. Instead of using raw frames via FrameZoom, the LLM interacts with CaptionZoom, enabling stronger, text-based visual reasoning. To enhance reasoning diversity, we set the LLM’s sampling temperature to 0.7 and generate 5 distinct reasoning trajectories for each input. The LLM iteratively performs reasoning and tool invocations until it reaches a final answer or the predefined step limit T . We retain only the trajectories whose predicted answer $\hat{y}_i$ matches the ground-truth y_i ; if no trajectory satisfies this condition, we randomly select one trajectory from the candidates. Through this process, we obtain diverse, interpretable, and temporally grounded reasoning traces that facilitate robust video reasoning performance.

3.3. Multimodal Tool-Reasoning Training

Based on the tool-augmented reasoning data generated by the LLM, we construct the video-interleaved CoTs dataset $\mathcal{D}_{\text{tool}} = \{(v_i, x_i, \hat{r}_i, \hat{y}_i)\}_{i=1}^M$. Each reasoning trajectory $\hat{r}_i$ consists of a sequence of reasoning steps and corresponding tool invocations. Among these, the FrameZoom tool replaces the CaptionZoom tool and plays a central role, as it is the only component capable of accessing the video content directly. It retrieves temporally localized visual evidence, providing the LLM with perceptual grounding essential for

Algorithm 2 Adaptive Reasoning in VideoThinker

- 1: **Input:** Video V ; duration L ; query Q ; confidence threshold τ ; number of sampled frames n ; VideoThinker p_θ
- 2: **Initialize:** retrieved clips $\mathcal{C} \leftarrow \emptyset$
- 3: **if** $L < 600s$ **then**
- 4: Sample frames $\{f_i\}_{i=1}^n \sim \text{Uniform}(V)$
- 5: $(\hat{y}, \gamma) \leftarrow p_\theta(\{f_i\}, Q)$ ▷ Direct reasoning
- 6: **else**
- 7: $\mathcal{C} \leftarrow \text{RetrieveTopKClips}(V, Q, k)$
- 8: $(\hat{y}, \gamma) \leftarrow p_\theta(\mathcal{C}, Q)$ ▷ Direct reasoning
- 9: **end if**
- 10: **if** $\gamma < \tau$ **then**
- 11: $(\hat{y}, \gamma) \leftarrow p_\theta(V, Q, \text{Prompt}_{\text{Tool.reason}})$ ▷ Tool reasoning
- 12: **end if**
- 13: **return** $\hat{y}$

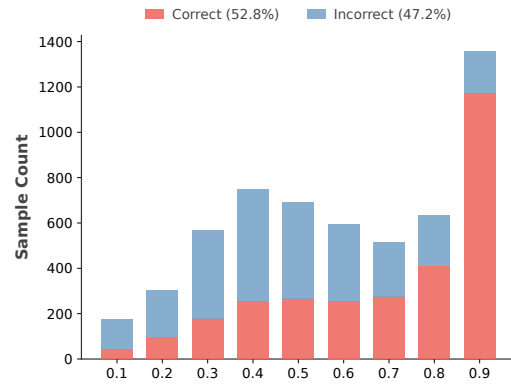


Figure 4. Confidence–accuracy relationship. The analysis is conducted on samples from VideoMME (2.7k), LongVideoBench (1.3k), and LVBench (1.5k).

reasoning. During dataset construction, the textual outputs of CaptionZoom are replaced by corresponding video segments represented as special `<video>` tokens. This conversion transforms the reasoning traces into multimodal interaction sequences, enabling direct visual-textual supervision for the VideoLLM. The multimodal student VideoLLM f_θ is trained to reproduce both the reasoning process and the final answer by minimizing the combined objective:

$$\mathcal{L} = \frac{1}{M} \sum_{i=1}^M \left[\ell(f_\theta^{(1:T)}(v_i, x_i), \hat{r}_i) \right], \quad (1)$$

where $\ell(\cdot)$ denotes the token-level cross-entropy loss. Through this training, the student VideoLLM learns to align textual reasoning with visual grounding, internalizing the teacher’s structured tool-usage logic. Consequently, it can perform interpretable, temporally grounded reasoning directly over video frames.

Model	#Frames	MLVU	LVBench	VideoMME (L)	LongVideoBench
Closed-source Models					
GPT-4o [12]	384	54.9	48.9	72.1	66.7
Gemini-1.5-pro [29]	256	–	33.1	77.4	64.0
Seed1.5VL-pro [28]	32	54.9	46.1	63.3	63.7
Open-source Models					
Qwen2.5VL-72B [1]	128	53.8	47.4	64.6	60.3
LongVILA-7B [4]	256	49.0	–	52.1	57.7
Video-XL-7B [26]	256	45.5	–	54.9	50.7
Video-R1-7B [8]	64	–	–	52.2	–
Qwen2.5VL-7B [1]	64	48.0	38.3	50.0	56.7
Agentic LLMs					
VideoAgent [33] (GPT-4)		52.2	-	46.2	-
VideoAgent [7] (GPT-4)		55.4	-	48.1	-
VideoTree [36] (Qwen-plus)		51.6	-	39.3	-
VideoExplorer [45] (Qwen2.5-7B-tuning + Qwen2.5-VL-32B)		58.6	51.4	-	-
Language Agent (Qwen-235B w/ our tools)		50.9	45.4	-	58.2
VideoThinker (Qwen2.5VL-7B w/ Think with videos)	64	54.8 (+6.8)	48.9 (+10.6)	53.7 (+3.7)	59.7 (+3.0)

Table 1. Evaluation results on MLVU, LVBench, VideoMME (L), and LongVideoBench. **VideoThinker** outperforms open-source models and demonstrates competitive performance with both closed-source models and LLM-based language agents.

3.4. Adaptive Video Reasoning

As shown in Figure 4, model confidence exhibits a strong correlation with prediction accuracy. Approximately 90% of the samples with confidence scores between 0.9 and 1.0 are correct, whereas more than half of those with confidence below 0.5 are incorrect. Motivated by this observation, we use confidence as a control signal to trigger multi-round tool-based reasoning. Given an input video and a query, VideoThinker first samples n frames if the video duration is shorter than 600 seconds; otherwise, it retrieves k relevant clips. The reasoning model $VideoThinker(p_\theta)$ then produces an initial answer along with a confidence score γ .

$$\gamma = \exp\left(\frac{1}{m} \sum_{t=1}^m \log f_\theta(\hat{y}_t \mid v, x, \hat{\mathbf{y}}_{<t})\right). \quad (2)$$

If $\gamma > \tau$, the answer is directly returned; otherwise, VideoThinker initiates a tool-augmented reasoning process to refine the response using the tool-reasoning prompt. This adaptive two-stage design effectively balances efficiency and accuracy by combining direct reasoning with deeper, tool-guided reasoning under uncertainty.

4. Experiments

4.1. Experimental Setup

Benchmarks To evaluate the performance of our model, we tested it on four distinct long video benchmarks: LongVideoBench [37], VideoMME [9], LVBench [32], and MLVU [57]. These benchmarks cover a variety of tasks

to assess the model’s video comprehension and multimodal reasoning capabilities. LongVideoBench [37] is a benchmark for long video understanding, consisting of 3,763 videos (up to 1 hour) with subtitles and 6,678 human-annotated multiple-choice QA pairs across 17 categories. We use the validation dataset for evaluation. VideoMME [9] contains 900 videos (totaling 254 hours) with 2,700 human-curated QA pairs across six domains and 30 subcategories. The videos range from 11 seconds to 1 hour and include frames, subtitles, and audio, enabling multimodal reasoning. We mainly evaluate using the subset of long videos (over 600 seconds). LVBench [32] focuses on extreme long video comprehension, with videos ranging from 70 seconds to 4 hours. It includes single-scene, multi-scene, and full-scene settings, covering diverse reasoning types such as temporal, spatial, causal, hypothetical, and external knowledge. MLVU [57] is a benchmark for multi-task long video understanding, featuring a diverse set of videos from genres like movies, surveillance, egocentric videos, and cartoons. We use the test set for evaluation.

Implementation Details We construct the multimodal video-interleaved tool reasoning CoTs based on CG-Bench [2], which includes 10k multiple-choice QA instances. The LLM agent uses Qwen3-235B-A22B-MoE [40] deployed on 4 NVIDIA H200 GPUs, while the VideoLLM is Qwen2.5-VL-7B [1]. The VideoLLM is fine-tuned on 10k video-interleaved samples for 3 epochs, utilizing up to 4 NVIDIA H200 GPUs. Training and inference are carried out using MS-Swift [55]. For efficient reasoning, Video-

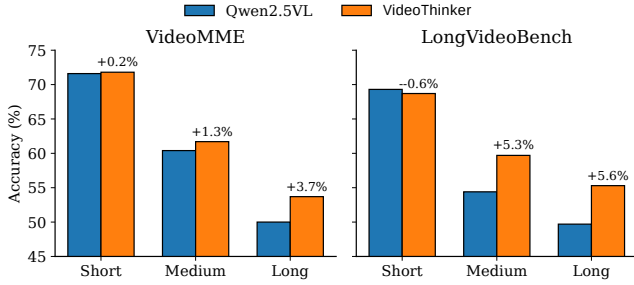


Figure 5. Accuracy on LongVideoBench and VideoMME across different video durations. Short: < 2 min; Medium: 2–15 min; Long: > 15 min.

Thinker uses 32 frames to directly generate the answer and its confidence, with each frame having a maximum resolution of 32,768 pixels. During the two-stage reasoning process, the total number of frames involved does not exceed 64. We evaluate VideoThinker against several baseline models, including closed-source and open-source models:

- **Closed-source models:** GPT-4o [12], Gemini-1.5 Pro [29], and Seed 1.5VL-Pro [28].
- **Open-source models:** Qwen2.5VL-72B [1], LongVILA-7B [4], Video-XL-7B [26], Video-R1-7B [8], and Qwen2.5VL-7B [1].
- **Agentic LLMs:** VideoAgent (with GPT-4) [33], VideoAgent (with GPT-4) [7], VideoTree (with Qwen-plus) [36], and VideoExplorer (Qwen2.5-7B-tuning + Qwen2.5-VL-32B) [45].

For the LVBench and MLVU benchmarks, audio subtitles are generated using FFmpeg and Whisper [25]. All decoding follows the official configurations, with the confidence threshold set to 0.7 by default.

4.2. Main results

As shown in Table 1, VideoThinker consistently outperforms open-source models and remains competitive with both closed-source systems and LLM-based agents. Specifically, VideoThinker achieves 54.8% on MLVU, surpassing Qwen2.5VL-7B (48.0%) and Qwen2.5VL-72B (53.8%), and performing on par with GPT-4o (54.9%). On LVBench, it reaches 48.9%, outperforming Qwen2.5VL-7B (38.3%) and Seed1.5VL-Pro (46.1%), and matching GPT-4o. For VideoMME, VideoThinker achieves 53.7%, exceeding Qwen2.5VL-7B (50.0%). On LongVideoBench, it obtains 59.7%, surpassing Qwen2.5VL-7B (56.7%) and closely approaching Qwen2.5VL-72B (60.3%). VideoExplorer achieves 58.6% on MLVU and 51.4% on LVBench, whereas VideoThinker obtains 54.8% and 48.9% with a single 7B model. In contrast, VideoExplorer is implemented as a language-agent pipeline that couples a separate large-scale LLM with an external Video-LLM, while VideoThinker is a single end-to-end 7B VideoLLM, highlighting the effi-

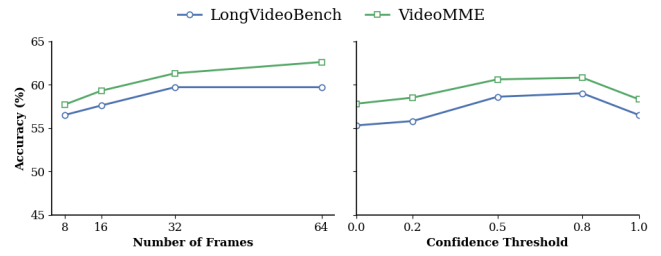


Figure 6. Performance of the VideoThinker at varying confidence threshold on Long-form VideoQA.

ciency of our approach. Notably, in the LLM-based agent setting, we compare against the language agents used in our data synthesis process. Compared with this redundant architecture—where VideoLLM acts as the tool, VideoThinker achieves an overall improvement of about 3%, including a +3.9% gain on MLVU. Overall, these results highlight VideoThinker’s strong performance across benchmarks, showing that our “Thinking with Videos” framework enables competitive long-form video reasoning while maintaining a lightweight design.

4.3. Ablation studies

Effect of Video Duration As shown in Figure 5, VideoThinker performs on par with Qwen2.5-VL when handling short videos (under 2 minutes). However, its advantage becomes evident as the video length increases. For medium videos (between 2 and 15 minutes), VideoThinker achieves consistently higher accuracy on both VideoMME and LongVideoBench, showing stronger capability in modeling longer temporal dependencies. The improvement is even more significant on long videos (exceeding 15 minutes), where VideoThinker surpasses Qwen2.5-VL by 3.7 percentage points on VideoMME and 5.6 percentage points on LongVideoBench. These results demonstrate that VideoThinker maintains stable reasoning and robust understanding in extended temporal contexts, effectively addressing the information loss and temporal drift commonly encountered in long-form video understanding.

Impact of n and τ on Performance We analyze the effects of the number of frames (n) and the confidence threshold (τ) on model performance. As shown in Figure 6, increasing n from 8 to 64 significantly improves accuracy—from 56.5% to 59.7% on LongVideoBench and from 57.7% to 62.6% on VideoMME—indicating that denser sampling provides richer visual context for reasoning. For the confidence threshold τ , which controls the switch between fast and tool-based reasoning, performance improves as τ increases from 0.0 to 0.8 (from 55.3% to

Dataset	Retrieval_clips (Top- <i>k</i>)				Retrieval_subtitles (Top- <i>k</i>)			
	1	3	5	10	1	3	5	10
LongVideoBench	59.7	59.8	58.8	58.6	57.4	58.4	58.0	57.1
VideoMME (L)	53.7	52.3	54.2	54.7	51.2	50.2	53.1	52.3
LVBench	48.9	49.1	47.2	49.0	50.6	50.5	49.0	49.3

Table 2. Ablation of **Retrieval_clips** and **Retrieval_subtitles** with varying Top-*k* values (1, 3, 5, 10) on three datasets.

Think with Videos

Question: In front of a light yellow wall, there is a woman wearing a pink hat sitting with another woman with long hair wearing a dark blue outfit. When the subtitle mentions 'the old people that can't work anymore,' what is the woman with the pink hat wearing?"
A. red short sleeves B. red long sleeves C. black long sleeves
D. pink short sleeves

Output:
To solve this question, I need to determine the exact moment in the video when the subtitle "the old people that can't work anymore" appears and then analyze the visual content at that time to identify what the woman with the pink hat is wearing.

Step 1: Locate the time interval when the subtitle "the old people that can't work anymore" appears.
I will use the 'Subtitle_Retrieval' tool to find the exact time interval when this subtitle appears.

```
<tool_call>
{
  "name": "subtitle_retrieval",
  "arguments": {
    "text": "the old people that can't work anymore"
  }
}
</tool_call>
```

<tool_use_result>
subtitle_retrieval
The interval: [(357.11, 357.12):
'the old people that can't work anymore
</tool_use_result>

The subtitle "the old people that can't work anymore" appears at approximately **357.11 to 357.12 seconds**. This is a very short interval, so I need to zoom in around this time to analyze the visual content accurately.

Step 2: Zoom into the interval around the subtitle to examine the scene. I will use the 'frame_zoom' tool to focus on a slightly broader window around 350–360 seconds to capture the visual details.

```
<tool_call>
{
  "name": "frame_zoom",
  "arguments": {
    "interval": [350, 360]
  }
}
</tool_call>
```

<tool_use_result>
frame_zoom

351 sec

354 sec

357 sec

360 sec

</tool_use_result>
The zoomed-in interval (350, 360) reveals that the woman with the pink hat is wearing a **red short sleeves** shirt. This directly answers the question about her clothing when the subtitle "the old people that can't work anymore" appears.

Final Answer:
A. red short sleeves

Figure 7. Case Study: VideoThinker performs agentic tool use by retrieving subtitles to locate temporal intervals and zooming into relevant frames for video reasoning.

59.0% on LongVideoBench and from 57.8% to 60.8% on VideoMME), but drops when $\tau = 1.0$ (56.5% and 58.3%, respectively). A low τ causes overreliance on direct reasoning, while a high τ triggers excessive tool reasoning. These results suggest that adaptive confidence-gated reasoning, combined with more frame samples, yields the best overall performance.

Ablation on Retrieval Tools We analyze the effect of retrieval by varying the number of retrieved clips and subtitles (Top-*k*). As shown in Table 2, Top-1 is generally suboptimal, indicating that a single retrieved item provides limited context. Increasing *k* typically improves performance as the model benefits from richer and more diverse information. On **LongVideoBench**, the best results appear at Top-3 for clips (59.8) and subtitles (58.4), showing a good balance between relevance and noise. **VideoMME (L)** favors broader context, peaking at Top-10 for clips (54.7) and Top-5 for subtitles (53.1). Top-1 is often slightly worse or comparable to larger *k*, though on LVBench subtitle retrieval, Top-1 performs best, indicating that the optimal *k* can be dataset-dependent.

4.4. Case Study

To further illustrate the multi-step reasoning process of VideoThinker, we present a representative case in Figure 7. The model is asked the question: “When the subtitle mentions ‘the old people that can’t work anymore,’ what is the woman with the pink hat wearing?” VideoThinker first invokes the Subtitle_Retrieval tool to locate the temporal position of the mentioned subtitle. The system accurately identifies the corresponding timestamp at 357.11s, which serves as a temporal anchor for subsequent reasoning. Based on the retrieved timestamp, VideoThinker employs the frame_zoom tools to examine the visual context within a narrow temporal window (350–360s). It focuses on frames depicting the woman with the pink hat and analyzes her appearance in detail. Through this fine-grained inspection, the model correctly determines that the woman is wearing a red short-sleeve shirt.

5. Conclusion

In this work, we present VideoThinker, an agentic VideoLLM designed for long-form video understanding through adaptive Temporal Retrieval and Temporal Zoom tool use. By leveraging temporal retrieval and temporal zoom, VideoThinker enables adaptive exploration of key video moments, allowing for dynamic reasoning over long videos. Trained on synthetic tool-interaction data generated through LLM-guided reasoning, VideoThinker learns to reason across long videos without requiring pre-existing long-form understanding. Extensive evaluations on multiple long-form video benchmarks demonstrate that VideoThinker consistently outperforms strong VideoLLM and LLM-agent baselines, validating the effectiveness of adaptive, tool-augmented reasoning for long-form video understanding.

section

Acknowledgments

This work was supported by the NSFC project (No. 62072399), the Zhejiang Provincial Natural Sci-

ence Foundation of China under Grant No. LZ23F020009, the Fundamental Research Funds for the Central Universities (No. S20240030), MoE Engineering Research Center of Digital Library, China Research Centre on Data and Knowledge for Engineering Sciences and Technology.

References

- [1] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025. 1, 3, 6, 7
- [2] Guo Chen, Yicheng Liu, Yifei Huang, Yuping He, Baoqi Pei, Jilan Xu, Yali Wang, Tong Lu, and Limin Wang. Cg-bench: Clue-grounded question answering benchmark for long video understanding. *arXiv preprint arXiv:2412.12075*, 2024. 6
- [3] Xiuyuan Chen, Yuan Lin, Yuchen Zhang, and Weiran Huang. Autoeval-video: An automatic benchmark for assessing large vision language models in open-ended video question answering. *arXiv preprint arXiv:2311.14906*, 2023. 1
- [4] Yukang Chen, Fuzhao Xue, Dacheng Li, Qinghao Hu, Ligeng Zhu, Xiuyu Li, Yunhao Fang, Haotian Tang, Shang Yang, Zhijian Liu, et al. Longvila: Scaling long-context visual language models for long videos. *arXiv preprint arXiv:2408.10188*, 2024. 6, 7
- [5] Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blstein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025. 1
- [6] Jingfeng Zhang Di Zhao, Hongsheng Hu, Philippe Fournier-Viger, Gillian Dobbie, and Yun Sing Koh. Balancing invariant and specific knowledge for domain generalization with online knowledge distillation. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*, pages 2440–2448, 2025. 3
- [7] Yue Fan, Xiaojian Ma, Rujie Wu, Yuntao Du, Jiaqi Li, Zhi Gao, and Qing Li. Videoagent: A memory-augmented multimodal agent for video understanding. In *European Conference on Computer Vision*, pages 75–92. Springer, 2024. 1, 2, 6, 7
- [8] Kaituo Feng, Kaixiong Gong, Bohao Li, Zonghao Guo, Yibing Wang, Tianshuo Peng, Junfei Wu, Xiaoying Zhang, Benyou Wang, and Xiangyu Yue. Video-rl: Reinforcing video reasoning in mllms. *arXiv preprint arXiv:2503.21776*, 2025. 6, 7
- [9] Chaoyou Fu, Yuhao Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. *arXiv preprint arXiv:2405.21075*, 2024. 1, 3, 6
- [10] Shenghao Fu, Qize Yang, Yuan-Ming Li, Xihan Wei, Xiaohua Xie, and Wei-Shi Zheng. Love-rl: Advancing long video understanding with an adaptive zoom-in mechanism via multi-step reasoning. *arXiv preprint arXiv:2509.24786*, 2025. 1
- [11] Xu Huang, Weiwen Liu, Xiaolong Chen, Xingmei Wang, Hao Wang, Defu Lian, Yasheng Wang, Ruiming Tang, and Enhong Chen. Understanding the planning of llm agents: A survey. *arXiv preprint arXiv:2402.02716*, 2024. 2
- [12] Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024. 6, 7
- [13] Xin Lai, Junyi Li, Wei Li, Tao Liu, Tianjian Li, and Hengshuang Zhao. Mini-o3: Scaling up reasoning patterns and interaction turns for visual search. *arXiv preprint arXiv:2509.07969*, 2025. 1, 3
- [14] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, et al. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024. 1
- [15] Chenglin Li, Qianglong Chen, Zhi Li, Feng Tao, and Yin Zhang. Vcbench: A controllable benchmark for symbolic and abstract challenges in video cognition. *arXiv preprint arXiv:2411.09105*, 2024. 1
- [16] Chenglin Li, Feng Han, Ruilin Li, Qianglong Chen, Jingqi Tong, Yin Zhang, Jiaqi Wang, et al. Adaptive fast-and-slow visual program reasoning for long-form videoqa. *arXiv preprint arXiv:2509.17743*, 2025. 2
- [17] KunChang Li, Yinan He, Yi Wang, Yizhuo Li, Wenhai Wang, Ping Luo, Yali Wang, Limin Wang, and Yu Qiao. Videochat: Chat-centric video understanding. *arXiv preprint arXiv:2305.06355*, 2023. 1, 3
- [18] Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, et al. Mvbench: A comprehensive multi-modal video understanding benchmark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22195–22206, 2024. 1
- [19] Bin Lin, Bin Zhu, Yang Ye, Munan Ning, Peng Jin, and Li Yuan. Video-llava: Learning united visual representation by alignment before projection. *arXiv preprint arXiv:2311.10122*, 2023. 1
- [20] Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, et al. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024. 1
- [21] Jiahao Meng, Xiangtai Li, Haochen Wang, Yue Tan, Tao Zhang, Lingdong Kong, Yunhai Tong, Anran Wang, Zhiyang Teng, Yujing Wang, and Zhuochen Wang. Open-o3 video: Grounded video reasoning with explicit spatio-temporal evidence. *arXiv preprint arXiv:2510.20579*, 2025. 1, 3
- [22] Munan Ning, Bin Zhu, Yujia Xie, Bin Lin, Jiayi Cui, Lu Yuan, Dongdong Chen, and Li Yuan. Video-bench: A comprehensive benchmark and toolkit for evaluating video-based large language models. *arXiv preprint arXiv:2311.16103*, 2023. 1
- [23] OpenAI. Introducing openai o3 and o4-mini. <https://openai.com/index/introducing-o3-and-o4-mini/>, 2025. Accessed: 2025-11-01. 1

- [24] Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, et al. Toolllm: Facilitating large language models to master 16000+ real-world apis. *arXiv preprint arXiv:2307.16789*, 2023. 2
- [25] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*, pages 28492–28518. PMLR, 2023. 4, 7
- [26] Yan Shu, Zheng Liu, Peitian Zhang, Minghao Qin, Junjie Zhou, Zhengyang Liang, Tiejun Huang, and Bo Zhao. Video-xl: Extra-long vision language model for hour-scale video understanding. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 26160–26169, 2025. 6, 7
- [27] Zhaochen Su, Linjie Li, Mingyang Song, Yunzhuo Hao, Zhengyuan Yang, Jun Zhang, Guanjie Chen, Jiawei Gu, Juntao Li, Xiaoye Qu, et al. Openthinking: Learning to think with images via visual tool reinforcement learning. *arXiv preprint arXiv:2505.08617*, 2025. 3
- [28] ByteDance Seed Team. Seed1.5-vl technical report. *arXiv preprint arXiv:2505.07062*, 2025. 6, 7
- [29] Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024. 6, 7
- [30] Dianyi Wang, Wei Song, Yikun Wang, Siyuan Wang, Kaicheng Yu, Zhongyu Wei, and Jiaqi Wang. Autoregressive semantic visual reconstruction helps vlms understand better. *arXiv preprint arXiv:2506.09040*, 2025. 3
- [31] Dawei Wang, Chengming Zhou, Di Zhao, Xinyuan Liu, Marci Chi Ma, Gary Ushaw, and Richard Davison. Towermind: A tower defence game learning environment and benchmark for llm as agents. *arXiv preprint arXiv:2601.05899*, 2026. 3
- [32] Weihang Wang, Zehai He, Wenyi Hong, Yean Cheng, Xiaohan Zhang, Ji Qi, Xiaotao Gu, Shiyu Huang, Bin Xu, Yuxiao Dong, et al. Lvbench: An extreme long video understanding benchmark. *arXiv preprint arXiv:2406.08035*, 2024. 1, 3, 6
- [33] Xiaohan Wang, Yuhui Zhang, Orr Zohar, and Serena Yeung-Levy. Videoagent: Long-form video understanding with large language model as agent. In *European Conference on Computer Vision*, pages 58–76. Springer, 2024. 1, 2, 3, 6, 7
- [34] Ying Wang, Yanlai Yang, and Mengye Ren. Lifelongmemory: Leveraging llms for answering queries in long-form egocentric videos. *arXiv preprint arXiv:2312.05269*, 2023. 2
- [35] Ye Wang, Qianglong Chen, Zejun Li, Siyuan Wang, Shijie Guo, Zhirui Zhang, and Zhongyu Wei. Simple o3: Towards interleaved vision-language reasoning. *arXiv preprint arXiv:2508.12109*, 2025. 1, 3
- [36] Ziyang Wang, Shoubin Yu, Elias Stengel-Eskin, Jaehong Yoon, Feng Cheng, Gedas Bertasius, and Mohit Bansal. Videotree: Adaptive tree-based video representation for llm reasoning on long videos. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 3272–3283, 2025. 1, 3, 6, 7
- [37] Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. Longvideobench: A benchmark for long-context interleaved video-language understanding. *Advances in Neural Information Processing Systems*, 37:28828–28857, 2024. 1, 3, 6
- [38] Junbin Xiao, Xindi Shang, Angela Yao, and Tat-Seng Chua. Next-qa: Next phase of question-answering to explaining temporal actions. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9777–9786, 2021. 3
- [39] Yuan Xie, Tianshui Chen, Zheng Ge, and Lionel Ni. Videomtr: Reinforced multi-turn reasoning for long video understanding. *arXiv preprint arXiv:2508.20478*, 2025. 3
- [40] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. 1, 4, 6
- [41] Junqi Yang, Yuecong Min, Jie Zhang, Shiguang Shan, and Xilin Chen. Infact: A diagnostic benchmark for induced faithfulness and factuality hallucinations in video-llms. *arXiv preprint arXiv:2603.11481*, 2026. 1
- [42] Zongxin Yang, Guikun Chen, Xiaodi Li, Wenguan Wang, and Yi Yang. Doraamongpt: Toward understanding dynamic scenes with large language models (exemplified as a video agent). *arXiv preprint arXiv:2401.08392*, 2024. 2
- [43] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *The eleventh international conference on learning representations*, 2022. 2
- [44] Zhou Yu, Dejing Xu, Jun Yu, Ting Yu, Zhou Zhao, Yueting Zhuang, and Dacheng Tao. Activitynet-qa: A dataset for understanding complex web videos via question answering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 9127–9134, 2019. 1
- [45] Huaying Yuan, Zheng Liu, Junjie Zhou, Hongjin Qian, Yan Shu, Nicu Sebe, Ji-Rong Wen, and Zhicheng Dou. Think with videos for agentic long-video understanding, 2025. 3, 6, 7
- [46] Huaying Yuan, Zheng Liu, Junjie Zhou, Ji-Rong Wen, and Zhicheng Dou. Videodeepresearch: Long video understanding with agentic tool using. *arXiv preprint arXiv:2506.10821*, 2025. 1, 2
- [47] Siyu Yuan, Kaitao Song, Jiangjie Chen, Xu Tan, Yongliang Shen, Ren Kan, Dongsheng Li, and Deqing Yang. Easy-tool: Enhancing llm-based agents with concise tool instruction. *arXiv preprint arXiv:2401.06201*, 2024. 2
- [48] Hang Zhang, Xin Li, and Lidong Bing. Video-llama: An instruction-tuned audio-visual language model for video understanding. *arXiv preprint arXiv:2306.02858*, 2023. 1
- [49] Haoji Zhang, Xin Gu, Jiawen Li, Chixiang Ma, Sule Bai, Chubin Zhang, Bowen Zhang, Zhichao Zhou, Dongliang He, and Yansong Tang. Thinking with videos: Multimodal tool-augmented reinforcement learning for long video reasoning. *arXiv preprint arXiv:2508.04416*, 2025. 1

- [50] Hao Zhang, Mengsi Lyu, Chenrui He, Yulong Ao, and Yonghua Lin. Trimtokenator: Towards adaptive visual token pruning for large multimodal models. *arXiv preprint arXiv:2509.00320*, 2025. [3](#)
- [51] Hao Zhang, Mengsi Lyu, Bo Huang, Yulong Ao, and Yonghua Lin. Trimtokenator-lc: Towards adaptive visual token pruning for large multimodal models with long contexts. *arXiv preprint arXiv:2512.22748*, 2025. [3](#)
- [52] Xiaoyi Zhang, Zhaoyang Jia, Zongyu Guo, Jiahao Li, Bin Li, Houqiang Li, and Yan Lu. Deep video discovery: Agentic search with tool use for long-form video understanding. *arXiv preprint arXiv:2505.18079*, 2025. [3](#)
- [53] Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. Video instruction tuning with synthetic data. *arXiv preprint arXiv:2410.02713*, 2024. [1](#)
- [54] Di Zhao, Yun Sing Koh, Gillian Dobbie, Hongsheng Hu, and Philippe Fournier-Viger. Symmetric self-paced learning for domain generalization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 16961–16969, 2024. [3](#)
- [55] Yuze Zhao, Jintao Huang, Jinghan Hu, Xingjun Wang, Yunlin Mao, Daoze Zhang, Zeyinzi Jiang, Zhikai Wu, Baole Ai, Ang Wang, Wenmeng Zhou, and Yingda Chen. Swift:a scalable lightweight infrastructure for fine-tuning, 2024. [6](#)
- [56] Ziwei Zheng, Michael Yang, Jack Hong, Chenxiao Zhao, Guohai Xu, Le Yang, Chao Shen, and Xing Yu. Deep-eyes: Incentivizing” thinking with images” via reinforcement learning. *arXiv preprint arXiv:2505.14362*, 2025. [1](#), [3](#)
- [57] Junjie Zhou, Yan Shu, Bo Zhao, Boya Wu, Zhengyang Liang, Shitao Xiao, Minghao Qin, Xi Yang, Yongping Xiong, Bo Zhang, et al. Mlvu: Benchmarking multi-task long video understanding. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 13691–13701, 2025. [6](#)
- [58] Wenrui Zhou, Mohamed Hendy, Shu Yang, Qingsong Yang, Zikun Guo, Yuyu Luo, Lijie Hu, and Di Wang. Flattery in motion: Benchmarking and analyzing sycophancy in video-llms, 2025. *arXiv:2506.07180 [cs.CL]*. [1](#), [3](#)
- [59] Bin Zhu, Bin Lin, Munan Ning, Yang Yan, Jiaxi Cui, Wang HongFa, Yatian Pang, Wenhao Jiang, Junwu Zhang, Zongwei Li, Cai Wan Zhang, Zhifeng Li, Wei Liu, and Li Yuan. Languagebind: Extending video-language pretraining to n-modality by language-based semantic alignment, 2023. [4](#)